Project-2

QuickBite | Food Delivery Operations Database

1. Company Background

QuickBite is one of India's fastest-growing food delivery platforms, connecting millions of hungry customers with thousands of restaurants across major metropolitan cities. Founded to make restaurant-quality food accessible at the tap of a button, QuickBite partners with diverse dining establishments — from neighbourhood Dhaba's and cloud kitchens to premium multi-cuisine restaurants — serving every palate and budget.

With hundreds of thousands of orders processed every month across cities such as Delhi, Mumbai, Bengaluru, Hyderabad, and Chennai, QuickBite operates a centralised analytics platform — QuickBite Analytics — to consolidate all delivery transactions, customer activity, restaurant performance, and agent operations. This empowers the leadership team to make fast, data-driven decisions on fleet management, restaurant partnerships, customer loyalty, and operational efficiency.

2. Business Scenario

The Operations & Strategy team at QuickBite is preparing for its quarterly business review. To support the review, the Head of Analytics has tasked the data team with producing a comprehensive analysis of the company's order history stored in the QuickBite Analytics database.

The analysis must address questions across four key business domains:

- Geographic & Revenue Performance — Understanding which cities drive orders and revenue, and where growth opportunities lie
- Restaurant & Cuisine Intelligence — Identifying top-performing restaurants, cuisine profitability benchmarks, and partner competitiveness
- Customer Behaviour — Profiling high-value customers, loyalty signals, spending trends, and year-on-year changes
- Delivery Operations & Fleet Management — Evaluating agent performance, SLA compliance, and city-level capacity planning

As the lead Data Analyst assigned to this project, you are required to write SQL queries against the QuickBite Analytics database to answer 21 questions raised by senior stakeholders — 15 business analysis questions and 6 data anomaly detection queries.

3. Database Schema — QuickBite Analytics

The database follows a Star Schema design with one central Fact table recording every delivery transaction, supported by three Dimension tables providing contextual details about customers, restaurants, and delivery agents.

DIM_RESTAURANT — Stores details of all partner restaurants

Column Name	Data Type	Key	Description
RestaurantID	INT	PK	Unique identifier for each restaurant
Restaurant_Name	VARCHAR(50)		Name of the restaurant
City	VARCHAR(30)		City where the restaurant is located
Cuisine_Type	VARCHAR(30)		Type of cuisine served (e.g., North Indian, Chinese)
Avg_Rating	DECIMAL(3,1)		Average customer rating out of 5.0
Opening_Year	INT		Year the restaurant was established

DIM_CUSTOMER — Stores customer profile information

Column Name	Data Type	Key	Description
CustomerID	INT	PK	Unique identifier for each customer
Customer_Name	VARCHAR(50)		Full name of the customer
City	VARCHAR(30)		City of residence
Email	VARCHAR(100)		Customer email address
Phone	VARCHAR(15)		Contact number
Member_Since	DATE		Date the customer first registered on the platform

DIM_DELIVERY_AGENT — Stores delivery agent profile information

Column Name	Data Type	Key	Description
AgentID	INT	PK	Unique identifier for each delivery agent
Agent_Name	VARCHAR(50)		Full name of the delivery agent
City	VARCHAR(30)		City the agent is assigned to
Vehicle	VARCHAR(20)		Type of delivery vehicle (Bike / Scooter)
Rating	DECIMAL(3,1)		Agent performance rating out of 5.0

FACT_ORDERS — Central fact table recording every delivery transaction

Column Name	Data Type	Key	Description
OrderID	INT	PK	Unique identifier for each order
CustomerID	INT	FK	Links to DIM_CUSTOMER
RestaurantID	INT	FK	Links to DIM_RESTAURANT
AgentID	INT	FK	Links to DIM_DELIVERY_AGENT
Order_Date	DATE		Date the order was placed
Order_Amount	DECIMAL(8,2)		Total bill amount for the order (INR ₹)
Delivery_Minutes	INT		Time taken to deliver the order (in minutes)
Rating	DECIMAL(3,1)		Customer rating given for this order (1.0–5.0)
Payment_Mode	VARCHAR(20)		Payment method (UPI / Card / Cash / Wallet)

4. Business Questions

The following 15 questions have been raised by QuickBite's senior leadership. Each question is accompanied by its business objective to help you understand the context and frame your analysis accordingly.

Q#	Business Question	Business Objective
Q1	Which cities recorded at least one order in 2022, and what were their total order counts and total revenue?	Identify geographic demand hotspots to guide city-level expansion strategy and marketing spend allocation.
Q2	Which cuisine type generates the highest total revenue and order volume across the platform?	Evaluate cuisine profitability to inform restaurant onboarding priorities and promotional partnerships.
Q3	What is each restaurant's total orders, total revenue, and average order value, ranked by total revenue?	Build a full restaurant performance league table to support partner tier classification and revenue negotiations.
Q4	What is the highest single order amount ever recorded per restaurant, along with the restaurant name and cuisine type?	Benchmark peak spend per restaurant to guide premium menu pricing strategy and upsell opportunities.
Q5	Who are the top 10 customers by lifetime spend, and what are their cities, total orders, and total amount spent?	Identify Gold Loyalty tier candidates for CRM targeting, exclusive offers, and personalised retention campaigns.
Q6	What is each delivery agent's total deliveries completed and average delivery time, ranked by total deliveries?	Rank delivery partner performance to surface top agents for incentive programmes and workload balancing.

Q7	What is the average delivery time per city, ordered from fastest to slowest?	Benchmark SLA compliance across cities to identify geographies requiring additional fleet resources.
Q8	Which restaurants received orders in 2023 but had zero orders in 2022?	Track newly onboarded or re-activated restaurant partners to measure platform growth and partner acquisition success.
Q9	Which customers placed more than 5 orders and maintained an average order value above ₹600?	Flag high-value engaged customers for premium loyalty rewards and priority customer success outreach.
Q10	What are the total revenue, total orders, and average order value for each year from 2020 to 2023?	Conduct a year-over-year revenue health check to track platform growth trends and set annual targets.
Q11	For each cuisine type, which restaurant holds the highest average order rating, and what are its total orders?	Identify best-rated restaurants per cuisine for featured placement, marketing spotlights, and annual awards.
Q12	What is the transaction count, total revenue, and percentage revenue share for each payment mode?	Understand payment mode adoption to guide negotiations with UPI, card, and wallet payment partners.
Q13	Which restaurant achieved the highest percentage revenue growth from 2022 to 2023?	Identify the most improved restaurant partner for the annual awards and to model replicable growth strategies.
Q14	For the 5 most active customers by total orders, what are their orders and revenue broken down by year (2020–2023)?	Analyse order frequency trends for top customers to detect churn risk and design personalised engagement plans.
Q15	For each city, how many delivery agents are assigned, and what is the total orders delivered and average orders per agent?	Conduct a city-level fleet capacity review to right-size agent deployment and optimise delivery coverage.

5. Data Anomaly Detection Questions (Q16 – Q21)

The datasets provided intentionally contain embedded data anomalies. The following 6 questions require you to write SQL queries that detect data quality issues, referential integrity violations, and fraud signals within the QuickBite Analytics database.

Q#	Business Question	Business Objective
Q16	Are there customers who share the same name and email address but have been assigned different CustomerIDs?	Detect duplicate customer accounts that may be artificially inflating the registered user base and skewing CRM metrics.
Q17	Did any delivery agent complete an unusually high number of orders (more than 50) on a single calendar day? Show the date and order count.	Flag potential fraud or system error — an agent completing 50+ orders in one day is operationally implausible and warrants investigation.

Q18	Which orders have an Order_Amount more than 3 times the platform-wide average order amount?	Identify potential billing errors or data entry outliers that may distort revenue reporting and financial reconciliation.
Q19	Which orders had a Delivery_Minutes value exceeding 180 minutes? Include the restaurant name, agent name, city, and delivery time.	Investigate severe SLA breaches — deliveries taking 3x the standard benchmark signal fleet failures, system issues, or data errors.
Q20	Are there any orders in Fact_Orders where the Order_Date is earlier than the customer's Member_Since date?	Verify referential integrity — an order placed before a customer registered is logically impossible and indicates a data consistency violation.
Q21	List all orders placed by CustomerIDs identified as duplicates (same name and email as another customer), showing order details and both the original and duplicate CustomerIDs.	Audit transactions linked to flagged duplicate accounts to assess the extent of orphan order records and their impact on revenue attribution.

6. Anomaly Reference Guide

The following anomalies have been deliberately embedded in the dataset. Use this as a reference to verify that your detection queries correctly surface each issue.

ID	Anomaly Type	Table Affected	Description
A1	**Duplicate Records**	DIM_CUSTOMER	5 customers duplicated with the same Name & Email but assigned different CustomerIDs (IDs 501–505)
A2	**Abnormal Activity**	FACT_ORDERS	AgentID 7 completed 85 orders on a single day (2023-03-15) — far above the daily operational average
A3	**Outlier Amounts**	FACT_ORDERS	20 orders with Order_Amount between ₹9,500 and ₹18,000 — approximately 6–8× the platform average
A4	**SLA Breach**	FACT_ORDERS	30 orders with Delivery_Minutes between 181–360 min — 3× the expected 60-minute delivery benchmark
A5	**Integrity Violation**	FACT_ORDERS	15 orders placed before the corresponding customer's Member_Since date — impossible in a consistent system
A6	**Orphan Transactions**	FACT_ORDERS	15 orders linked to duplicate CustomerIDs (501–505) which should not exist in a clean, deduplicated system